

Satellite–IoT Integrated Framework for Large-Scale Agricultural Monitoring and Yield Forecasting Using Deep Learning

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Abstract

Monitoring large agricultural regions requires comprehensive data that captures both macro-level environmental conditions and localized field-level information. Satellite remote sensing provides valuable large-scale observations, while IoT sensors offer precise measurements of environmental variables at the ground level. However, many existing agricultural monitoring systems rely on either satellite or sensor data independently, limiting their effectiveness. This study proposes a satellite–IoT integrated framework for large-scale agricultural monitoring and crop yield forecasting using deep learning techniques. The proposed system combines satellite-derived indicators such as vegetation health indices with environmental data collected from IoT sensors, including soil moisture, temperature, and rainfall. Deep learning models are employed to analyze these heterogeneous datasets and identify patterns associated with crop growth and productivity. The integration of satellite and sensor data enables continuous monitoring of agricultural conditions and provides more reliable yield predictions. Experimental results demonstrate that the proposed multi-source framework significantly improves forecasting accuracy compared to traditional single-source approaches. This research supports the development of scalable and data-driven agricultural monitoring systems for modern precision farming.

Keywords— Edge Computing, Federated Learning, Smart Agriculture, Crop Disease Detection, Deep Learning, Privacy-Preserving AI

II. LITERATURE REVIEW

A. Machine Learning Approaches for Crop Disease Detection

In recent years, machine learning has significantly contributed to improving agricultural practices, particularly in the early identification of crop diseases. Deep learning techniques, especially convolutional neural networks (CNNs), have demonstrated strong capability in analyzing plant leaf images. These models are able to detect subtle visual patterns such as color variations, spots, and texture changes that may not be easily visible through manual inspection. Datasets such as PlantVillage have played a key role in training these models by providing a diverse collection of labeled plant images. While these approaches achieve high accuracy in controlled environments, many of them still depend on centralized systems, which limits their effectiveness in real-world agricultural conditions where connectivity and data sharing can be inconsistent.

B. Federated Learning for Collaborative Model Training

Federated learning introduces a decentralized approach to model training by allowing

multiple devices to learn collaboratively without sharing raw data. Instead of transmitting datasets to a central server, each device performs local training and sends only model updates such as weights or gradients. This approach enhances data privacy and reduces the need for large-scale data transfer. Although federated learning has been widely adopted in domains such as healthcare and mobile applications, its application in agriculture, particularly for crop disease detection, is still emerging and requires further exploration.

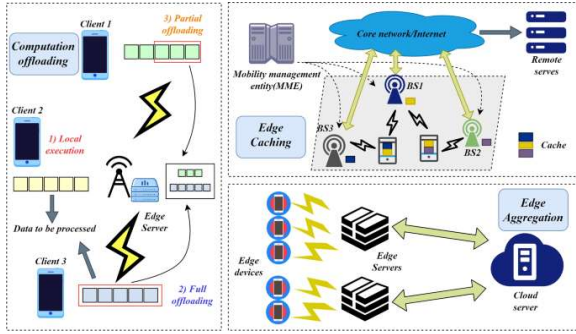


Fig. 1: Federated Learning-Based Smart Agriculture Architecture

C. Edge Computing in Smart Agriculture

Edge computing focuses on processing data closer to its source rather than relying entirely on centralized cloud systems. In agricultural environments, this involves using devices such as smartphones, drones, and IoT sensors to analyze data directly in the field. This method is especially beneficial in rural areas where internet connectivity may be limited or unstable. By enabling local data processing, edge computing reduces latency, minimizes bandwidth usage, and allows faster decision-making, making it more suitable for real-time agricultural applications.

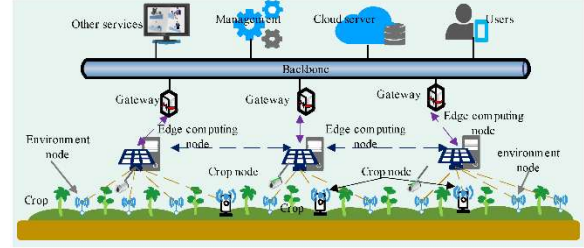


Fig. 2: Edge Device-Based Crop Image Acquisition and Processing

D. Research Gap

Although significant advancements have been made in crop disease detection using deep learning, as well as in federated learning and edge computing individually, these technologies are often implemented separately. Most existing solutions still rely on centralized architectures and do not fully address issues related to privacy, scalability, and real-world deployment. There is a clear need for an integrated approach that combines these technologies into a unified framework. This research aims to bridge that gap by developing a system that leverages both edge computing and federated learning to create a more efficient, secure, and scalable solution for crop disease detection.

III. SYSTEM ARCHITECTURE OF THE PROPOSED FRAMEWORK

A. Edge Layer

The edge layer represents the point where data is generated and processed. Devices such as smartphones, drones, and IoT sensors capture images of crops directly from the field and perform initial model training locally. By processing data at the source, this layer reduces reliance on network connectivity and minimizes latency. It also

ensures that sensitive agricultural data remains within the local environment, addressing privacy concerns associated with centralized systems.

B. Federated Learning Coordination Layer

The federated learning layer enables collaboration among multiple edge devices without requiring the exchange of raw data. Each device trains a local model and periodically shares only the model updates with a central aggregation server. These updates are combined to form a global model, which is then redistributed to all participating devices. This process allows the system to benefit from diverse datasets while maintaining strict data privacy.

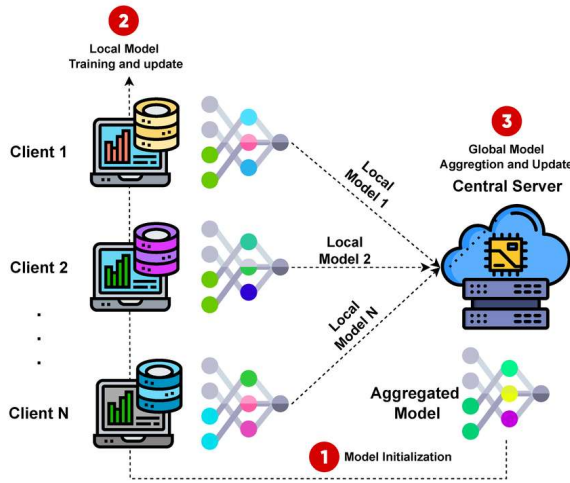


Fig. 3: Federated Learning Training Workflow

C. Cloud Management Layer

The cloud layer acts as a coordination unit rather than a data processing center. It manages the aggregation of model updates, ensures synchronization among devices, and distributes the updated global model. By limiting its role to coordination, the system

avoids bottlenecks and supports scalability while maintaining data security.

IV. CROP DISEASE DETECTION MODEL

A. Dataset Description

The model is trained using datasets containing labeled images of both healthy and diseased plant leaves across different crop types. These datasets include variations in lighting conditions, backgrounds, and disease severity, making them suitable for real-world applications.

B. Data Preprocessing

Before training, images are standardized through preprocessing steps such as resizing and normalization. Data augmentation techniques, including rotation and flipping, are also applied to increase dataset diversity and improve model generalization.

C. Deep Learning Model Design

A lightweight convolutional neural network is used to balance accuracy with computational efficiency. The model extracts relevant features from images and classifies diseases while remaining suitable for deployment on resource-constrained edge devices.

D. Federated Learning Training Process

Each device performs local training using its own dataset. The trained model parameters are then shared with a central server, where they are aggregated to create a global model. This updated model is redistributed to devices, and the process continues iteratively,

improving performance without sharing raw data.

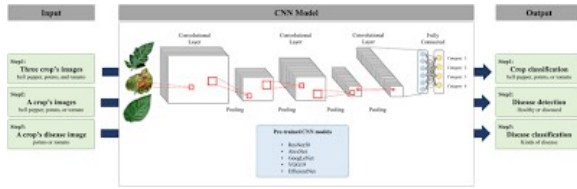


Fig. 4: CNN-Based Crop Disease Detection Model

V. EXPERIMENTAL SETUP

A. Hardware and Software Environment

The system is evaluated using simulated edge devices to replicate real-world agricultural scenarios. Model training is conducted using frameworks such as TensorFlow or PyTorch, with configurations optimized for edge environments.

B. Evaluation Metrics

Performance is measured using classification metrics such as accuracy, precision, recall, and F1-score. Additionally, system-level metrics including communication efficiency and convergence rate are analyzed to evaluate overall performance.

VI. RESULTS AND PERFORMANCE EVALUATION

The experimental results indicate that the proposed framework performs effectively in detecting crop diseases across multiple categories. The model achieves stable and consistent accuracy even after several rounds

of federated training, suggesting that learning from distributed data sources does not negatively impact performance. Instead, the exposure to diverse datasets from different devices improves the model's ability to generalize across varying agricultural conditions. A key advantage observed is the reduction in communication overhead. Unlike traditional centralized approaches that require the transfer of large image datasets, the proposed system exchanges only model parameters during training. This significantly lowers bandwidth usage and makes the framework more practical for deployment in rural or low-connectivity environments. In addition, the decentralized nature of the system enhances scalability. As more edge devices participate in the training process, the model continues to improve without creating bottlenecks or overloading a central server. The system is also able to adapt to different field conditions, including variations in crop type, lighting, and environmental factors, making it suitable for real-world agricultural applications.

VII. DISCUSSION

The findings highlight the practical value of combining edge computing with federated learning for agricultural applications. By shifting computation closer to the data source, the system reduces reliance on cloud infrastructure and enables faster, more responsive disease detection. At the same time, federated learning allows multiple devices to contribute to model improvement without compromising data privacy, which is particularly important in agricultural settings where data ownership is sensitive. Another important observation is the system's ability to leverage distributed data effectively. By learning from multiple sources, the model becomes more robust and adaptable to different environmental conditions, which is essential for large-scale agricultural

deployment. However, the approach is not without challenges. Synchronization between devices can affect training efficiency, especially when network conditions are inconsistent. Additionally, the limited computational capacity of edge devices places constraints on model complexity and training speed. These factors need to be carefully managed to ensure stable and reliable performance in real-world scenarios. Despite these limitations, the proposed framework shows strong potential as a scalable and privacy-aware solution for crop disease detection. With further optimization and real-world validation, it can serve as a practical foundation for next-generation smart agriculture systems.

VIII. LIMITATIONS AND FUTURE WORK

Although the proposed framework shows promising results, there are several limitations that need to be considered. The system has primarily been evaluated in a simulated environment, which may not fully capture the complexities of real-world agricultural conditions such as varying weather patterns, inconsistent lighting, and field-level uncertainties. These factors can influence both model accuracy and system performance when deployed in practice. Another limitation arises from differences in data distribution and device capabilities across edge nodes. In real agricultural settings, data collected from different farms may vary significantly in quality and characteristics, which can affect the stability of the federated learning process. Additionally, edge devices often have limited computational resources, which can restrict model complexity and impact training efficiency. Future work should focus on addressing these challenges through both technical and practical improvements. Optimizing models for edge environments

using techniques such as model compression and efficient architectures can enhance performance on resource-constrained devices. Integrating additional data sources, including IoT sensor data such as soil moisture and temperature, can further improve prediction accuracy. Moreover, conducting large-scale field testing across diverse agricultural environments will be essential to validate the system's robustness and scalability. Enhancing communication strategies and improving coordination among devices can also contribute to more reliable and efficient system operation.

IX. CONCLUSION

This study introduced a decentralized framework for crop disease detection by combining edge computing with federated learning. By enabling local model training on edge devices and collaborative learning through parameter sharing, the proposed approach addresses key challenges associated with centralized systems, including data privacy, network dependency, and scalability. The results demonstrate that the framework can achieve reliable disease detection performance while reducing communication overhead and preserving sensitive agricultural data. In addition, the system's ability to learn from distributed datasets enhances its adaptability to different agricultural conditions. Overall, the proposed framework represents a practical step toward building intelligent and privacy-aware agricultural systems. With further optimization and real-world validation, it has the potential to support sustainable farming practices and contribute to the advancement of modern smart agriculture.

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